Node Classification with GNNs: GCN vs GraphSAGE

Arian Madani • Amirhossein Sadeghi

Supervisor: Dr. Zeinab Maleki

Project Overview

- Problem: node classification on citation graphs. Each node is a paper; edges are citations.
- Inputs: sparse bag-of-words features (X) + graph connectivity (A).
- Goal: predict the research topic/class of each paper using both X and graph structure.
- Models: compare GCN vs GraphSAGE with the SAME depth (2 message-passing layers) + linear classifier.
- Training: cross-entropy on train nodes → pick best checkpoint via validation accuracy → report test metrics.
- Motivation: study how clustering, sparsity, and hubs influence GNN performance.

Model Architecture (clean text inside)

GCN — 2 layers

Input features X
↓
GCN Layer 1
↓
ReLU + Dropout
↓
GCN Layer 2
↓
Linear classifier
↓
Softmax → class

Uses ALL neighbors (normalized…

GraphSAGE — 2 layers

Input features X
↓
Sample neighbors (L1)
↓
SAGE Layer 1 (mean agg + concat)
↓
ReLU + Dropout
↓
Sample neighbors (L2)
↓
SAGE Layer 2 (mean agg + concat)
↓
Linear classifier
↓
Softmax → class

Uses sampled neighbors…

Datasets + Graph Structure

Planetoid citation datasets: nodes are scientific papers, edges are citations, and each node label is the paper's research topic/class.

Cora

Nodes: 2708 Edges: 10556
Features: 1433 Classes: 7

Cora is a citation network of machine learning papers. Each node is a paper and each edge indicates a citation. Labels correspond to 7 ML topic areas (e.g., Neural Nets, Probabilistic,…

CiteSeer

Nodes: 3327 Edges: 9104
Features: 3703 Classes: 6

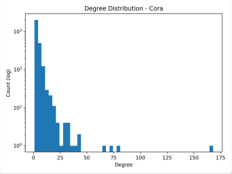
CiteSeer is a citation network of scientific papers. It is typically noisier than Cora, with citations across multiple CS topics. Labels correspond to 6 categories (Agents, AI, DB, IR, ML, HCI).

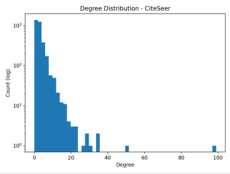
PubMed

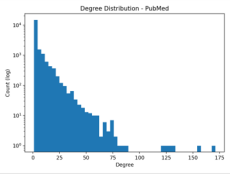
Nodes: 19717 Edges: 88648
Features: 500 Classes: 3

PubMed is a biomedical citation network focused on diabetes. It is larger and often sparser. Labels correspond to 3 diabetes-related classes (Experimental, Type 1, Type 2).

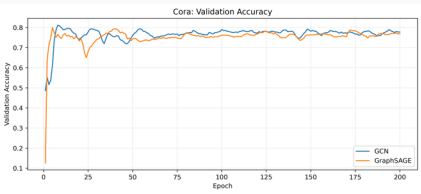
Degree Distributions (connectivity patterns):



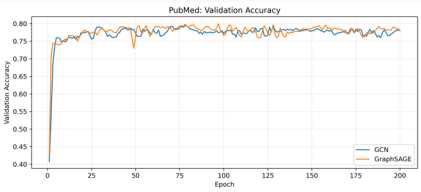




Validation Accuracy



Cora



PubMed

Key Results (Test)

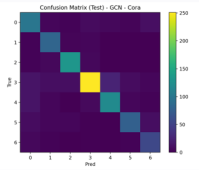
Cora:
GCN Acc 81.5% | Macro-F1 0.804 | 5.10s
GraphSAGE Acc 79.4% | Macro-F1 0.782 | 11.81s

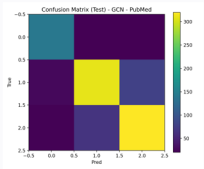
PubMed:
GCN Acc 76.9% | Macro-F1 0.767 | 21.93s
GraphSAGE Acc 76.3% | Macro-F1 0.760 | 31.00s

Conclusion

- Overall, GCN performs best on Cora & PubMed under the same 2-layer setup.
- Why: these graphs are relatively small and have meaningful local neighborhoods → full-neighbor aggregation helps.
- GraphSAGE remains attractive for very large graphs where sampling is necessary for efficiency.…

Confusion matrices (GCN): diagonal = correct predictions





Poster v9 • fixed architecture overflow